

Lecture 08 September 2026

Design for Values and AI

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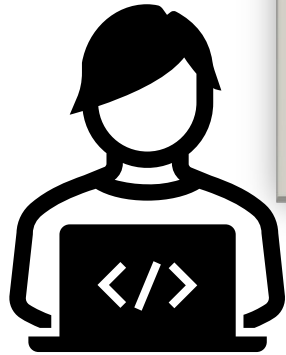
Lecture Overview

- Technology ethics: Design for Values and Value Sensitive Design
- Values and technology
- Some examples
- Applying DfV in your assignment

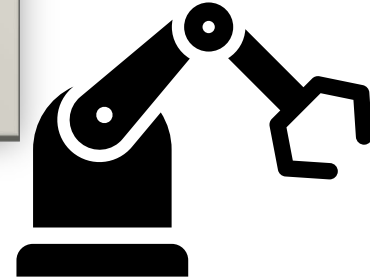
Where is the ethics?



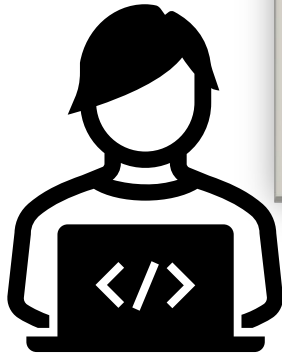
Two locations of ethics in engineering



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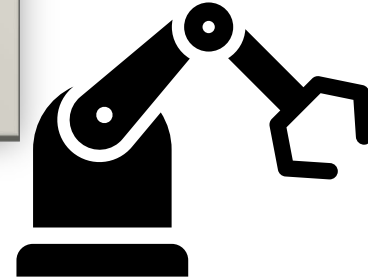
Two locations of ethics in engineering



The engineer



<https://www.bbc.com/audio/play/p0mh56kj>



The technology

Some basic tools of ethical analysis



Values



Norms



Virtues

What *ethical* values are at stake for AI?

Human agency and oversight

Technical robustness and safety

Privacy and data governance

Transparency

Diversity, non-discrimination and fairness

Societal and environmental wellbeing

Accountability

Design for Values

- In CHLEP Ch. 3, Design for Values and Value Sensitive Design are taken as equivalent (variations of a single type of approach).
- Design for values is an approach to incorporating ethical values into a technological design, in which there is input from:
 - Stakeholders
 - Ethical theories, esp. mid-level ethical theories

primary function

- The primary function of an artifact *A* is that function which defines what it is, and which explains what makes it *a good A* when it works correctly.

Ex: Paperweight, hammer, gun

This notion of “good” seems to contain no ethical value. Can’t we talk about a good bomb in this sense?

Do artifacts have politics?

- Langdon Winner (1980)

"Still life with onions and tomatoes" by Catherine M. Wood. Image in the public domain, via Wikimedia Commons

https://commons.wikimedia.org/wiki/File:Catherine_M._Wood_Still_life_with_onions_and_tomatoes_1891.jpg



the beaches of Long Island, New York



Robert Moses' bridges

- <http://www.nyc-architecture.com/BKN/BKN001.htm>



*example: Robert
Moses' bridges*

http://media.komonews.com/images/080417_bus_crash_3.jpg



http://en.wikipedia.org/wiki/File:Robert_Moses_with_Battery_Bridge_model.jpg



Another example from Winner



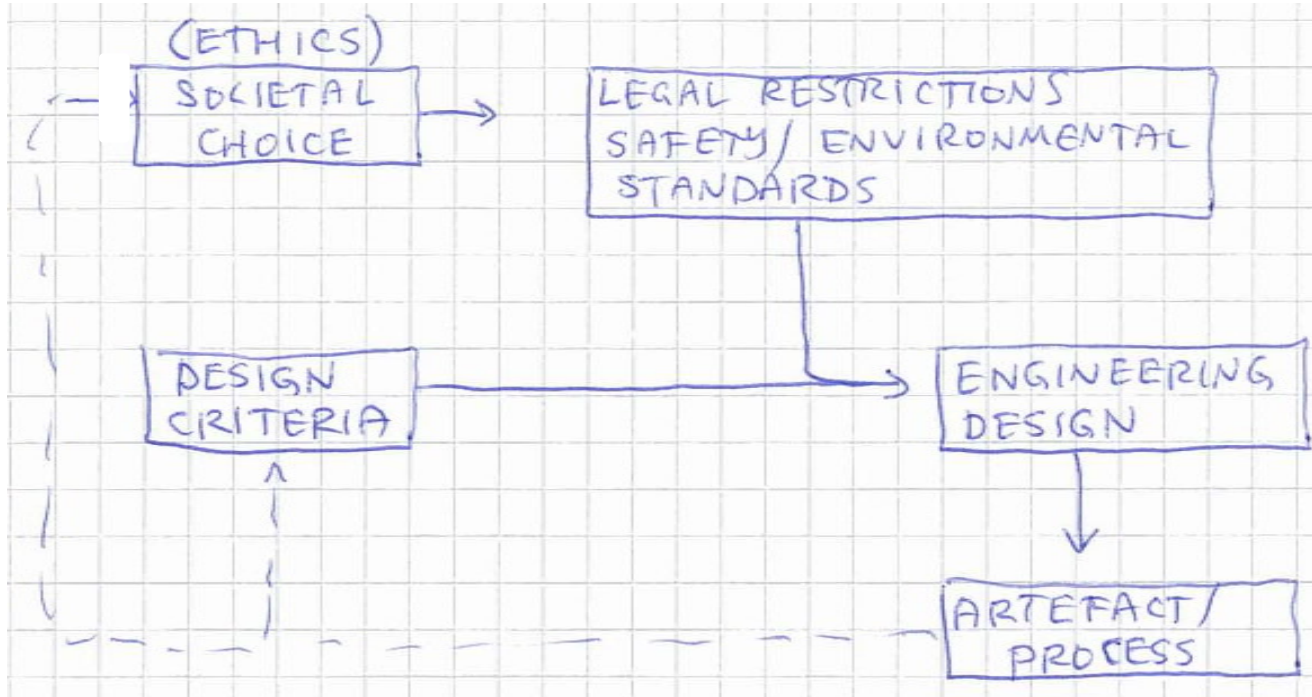
http://en.wikipedia.org/wiki/File:CTM_Johnson_Tomato_Harvester.JPG



Winner's main claims

- Technology is sometimes expressly created for a political reason that is logically and temporally prior to its primary function.
- Some technologies are inherently political in that their primary function (a) cannot be realized in practice without a particular political relationship, or (b) are much more difficult to realize in practice without a particular political relationship.

How do values get into technology?



ethics and design

Suppose we want to incorporate ethics into technology design. What do we need?

- (a method)
- knowledge of relevant ethical norms
- information about the technology and its likely effects on people
- information about society

domain-specificity of ethical judgment

- We often need to know how welfare goods are promoted, what moral rights are relevant, or what standards are relevant *in a given area*, in order to make an ethical judgment.
- For example:
 - chemicals: environmental and health norms
 - electronic voting: political rights
 - assisted fertility technology: reproductive rights/ norms of medical ethics
 - military robots: ethics of justified warfare, international legal conventions

Design for Values: sources of values

- DfV differs from VSD in its emphasis on theory as a source of values.
- General ethical theories (utilitarianism, deontology, and virtue ethics) provide “salience lenses” that allow us to identify some issues as ethically significant.
- “Mid-level” theories (Rawls’ theory of justice as fairness, the Capability Approach) provide more specific guidance about values to include. (Here, I would also turn to foundational sources such as the UDHR and the Belmont Report.)
- Participant and stakeholder values are also included, but in a qualified way.

The challenge of AI in Design for Values

Some of AI's “disruptive” features create special challenges for ethics, and for Design-for-Values:

- It is being adopted very quickly.
- It is applied at a massive scale.
- It can take intelligent steps independent of human judgment and intervention.
- It is often opaque, i.e., the process by which it takes steps cannot be recovered easily in the case of deep learning or LLMs.

Some methodological assumptions of DfV

- It is both possible and necessary to take steps early on in order to ensure that ethical values are built into our technological solutions.
- Each technology, considered in a narrow sense, can be embedded in social practices in different ways to create different socio-technical systems (“applications,” technologies in the broad sense).
- Our values can change over time.

Design for Values

- Design for Values is described as having three aspects (CHLEP p. 72):
 1. Identifying the relevant values
 2. Embedding these values in systems
 3. Assessing whether these efforts have been successful

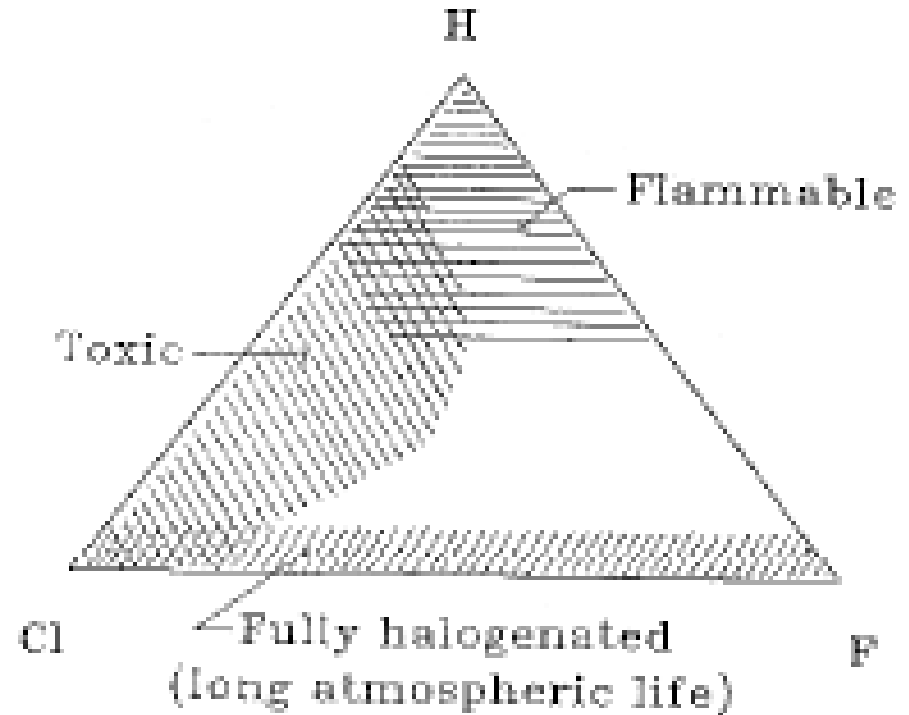
Classic case: a new refrigerant

interesting features of the desired artifact:

value trade-offs:

- patentability vs. affordability
- flammability vs. environmental harmfulness (depending on number of ozone-depleting Cl atoms in compound and number of greenhouse-gas F atoms in the compound)
- toxicity vs. flammability

diagram of ethical tradeoffs in refrigerants



Ibo van de Poel, Investigating Ethical Issues in Engineering Design, *Science and Engineering Ethics* 7, 3 (2001): 429-446.

AI applications: some approaches

1. Build the system using ethical rules/ values.
 - “Whenever you detect a human, stop the robot” [SAFETY]
 - “Do not make more systematic errors for any sub-group in the dataset compared to the others” [FAIRNESS]
2. Train systems on large data sets in which the rules are followed (e.g., cases where the robot stops for humans)
3. Train systems on large data sets in which the rules might not have been followed but impose the rules as filters.
4. Rely on the socio-technical system (users) to provide ethical input.

Cases from CHLEP Ch. 3

- Generative language models
- AI for determining loan eligibility
- AI for benefits fraud detection

Please look carefully at how the authors describe these cases 😊

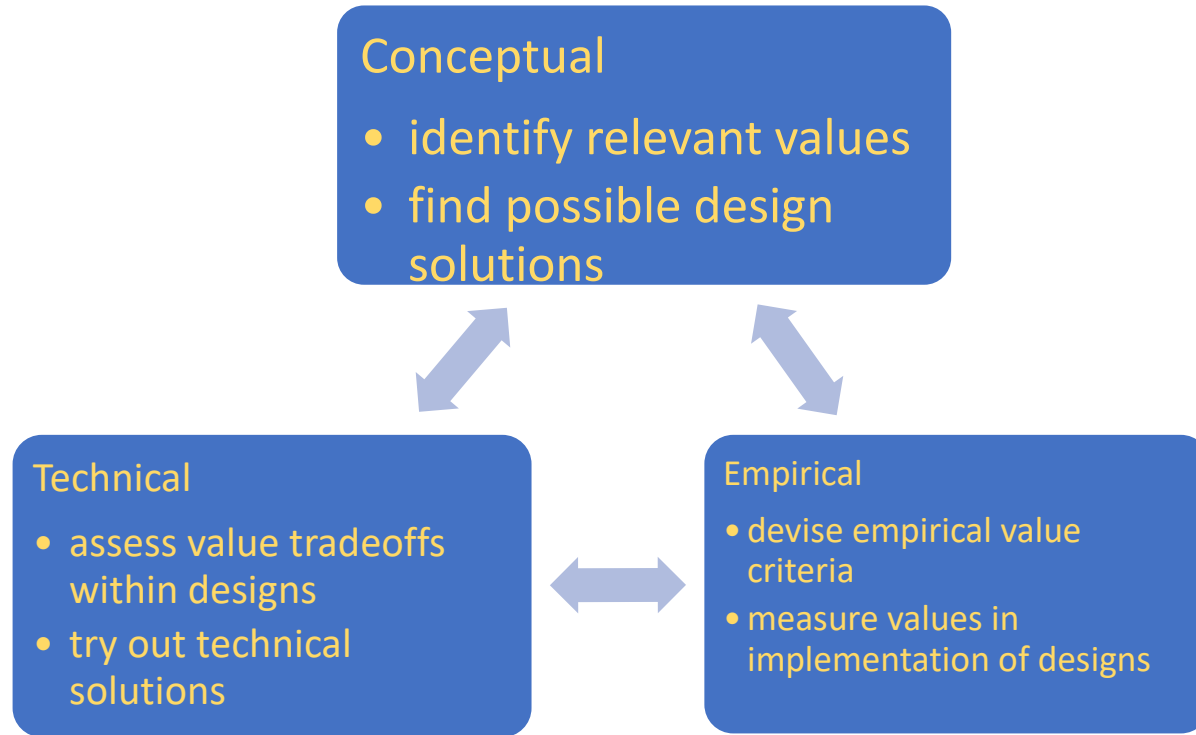
Source: Friedman, Kahn & Borning 2006.



Value Sensitive Design

Approach developed by Batya Friedman and colleagues ~ 20 years ago to integrate ethical values systematically in engineering design.

Value Sensitive Design



Where do the values in VSD come from?

List from Friedman et al. 2005:

human welfare

ownership & property

privacy

freedom from bias

universal usability

trust

courtesy

autonomy

informed consent

accountability

calmness

identity

environmental sustainability

Value Sensitive Design

- What values are relevant? Are there tradeoffs? (Conceptual)
- Are there technical solutions to value tradeoffs? (Technical)
- How do we measure the success of these solutions? (Empirical)

Source: Friedman,
Kahn & Borning
2006.

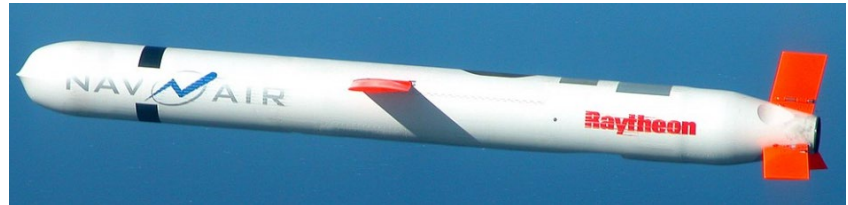


Figure 2. Plasma Display Technology Studies

the tactical Tomahawk

“The Tomahawk missile is the U.S. Navy’s premier land attack missile ... once launched, [it] cannot be redirected in-flight. This limitation to the system causes potential redundant demolition of targets, thus wasting the \$1.2 million dollar [sic] missile, as well as the inability to correct for a targeting mistake in flight. In addition ... the missile cannot respond to dynamic situations in which, for example, a target has moved, or a more critical target emerges.”

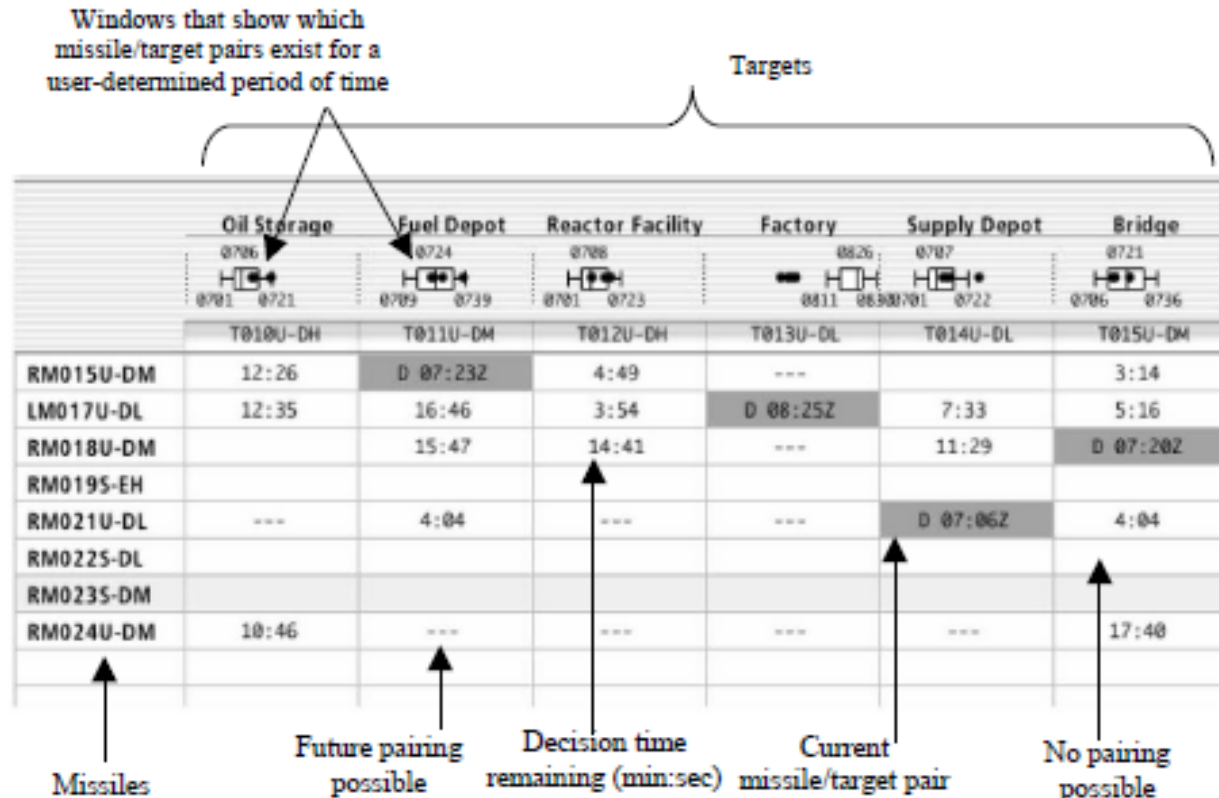
→ develop Tactical Tomahawk with re-direction ability



the role of ethics?

“The primary purpose of analyzing conceptual ethical issues in the context of a military command and control system is to identify human value design considerations that will guide the engineer in constructing a more effective decision support system. For example, understanding the principle of discrimination can guide the engineer in developing a system that accounts for the risk of harm to both friendly and non-combatant forces” (Cummings 2006, 706).

decision-support tool



level of automation

Automation Level	Automation Description
1	The computer offers no assistance: human must take all decisions and actions.
2	The computer offers a complete set of decision/action alternatives, or
3	narrows the selection down to a few, or
4	suggests one alternative, and
5	executes that suggestion if the human approves, or
6	allows the human a restricted time to veto before automatic execution, or
7	executes automatically, then necessarily informs humans, and
8	informs the human only if asked, or
9	informs the human only if it, the computer, decides to.
10	The computer decides everything and acts autonomously, ignoring the human.

In the empirical phase, what emerged?

---Automation bias

- Human operators have a tendency to accept the default of automated, “smart” systems. When the selection system for the missile highlighted a target, operators tended to accept that target, even if they possessed other information that should have led them to double-check it.
- This does not directly test well-being, but it does suggest a problem for the principle of human autonomy, and maybe for responsibility.

Some limitations of Design for Values

- Some ethical issues appear too late in the design process to be incorporated into the design.
- How people really use a technology is difficult to predict and introduces new ethical issues.
- In cases of high moral uncertainty ('moral disruption'), it may be impossible to specify values or weigh values against one another in the way required by Design for Values.
- Design for Values is very resource-intensive, especially if it has to be specified for each application.

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